

ELITE MATHEMATICS

Pearson Edexcel International A Level

Mechanics 1

Adaptive Classified

Chapter 07 • Newton's Second Law

With Answers

Questions with full worked solutions

- 58 pages
- Past papers up to June 2026

MECHANICS 1 | TOPIC 07

Newton's Second Law

Your classified route

25

QUESTIONS

287

MARKS

4

FAMILIES

10

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Read the diagram, choose the model, then use the working space to complete the mechanics.

TOPIC 07 | FAMILY 01

Newton's second law for one particle

WHAT THIS FAMILY ASKS

Resolve the resultant force and use $F = ma$ for a single particle, vehicle or lift.

VARIANT ROUTE**V01** Scalar $F = ma$ **V02** Lift or apparent reaction**6 classified questions**

44 total marks

Summer 2023 Q3 — complete original question

3. Two students observe a book of mass 0.2 kg fall vertically from rest from a shelf that is 1.5 m above the floor.

Student A suggests that the book is modelled as a particle falling freely under gravity.

- (a) Use student A 's model to find the time taken for the book to reach the floor.

(3)

Student B suggests an improved model where the book is modelled as a particle experiencing a constant resistance to motion of magnitude R newtons.

Given that the time taken for the book to reach the floor is 0.6 seconds,

- (b) use student B 's model to find the value of R

(5)

DR ESLAM AHMED

Summer 2023 Q3 — detailed solutionScalar $F = ma$ **ALL**
8 marks**Use free fall for student A**

From rest, with downward displacement 1.5 m and acceleration g , use $s = ut + \frac{1}{2}at^2$.

Find the time

$1.5 = 0 + \frac{1}{2}(9.8)t^2$, so $t^2 = \frac{3}{9.8}$ and $t = 0.553 \dots$ s.

Find the acceleration in student B's model

Now the time is 0.6 s. Again starting from rest, $1.5 = \frac{1}{2}a(0.6)^2 = 0.18a$, so $a = 8.333 \dots \text{ m s}^{-2}$ downward.

Apply Newtons second law

Taking downward as positive, weight is $0.2g$ and resistance R acts upward. Therefore $0.2g - R = 0.2a$.

Find R

$1.96 - R = 0.2(8.333 \dots) = 1.666 \dots$, so $R = 0.293 \dots$ N.

Final answer (a) $t = 0.553$ s approximately. (b) $R = 0.293$ N.

October 2023 Q3 — complete original question

3. A hammer is used to hit a tent peg into soft ground.

The hammer has mass 1.8 kg and the tent peg has mass 0.2 kg .

The hammer and tent peg are both modelled as particles and the impact is modelled as a direct collision.

Immediately before the impact, the tent peg is stationary and the hammer is moving vertically downwards with speed 10 m s^{-1}

Immediately after the impact, the hammer and tent peg move together, vertically downwards, with the **same** speed $v\text{ m s}^{-1}$

- (a) Find the value of v (2)
- (b) Find the magnitude of the impulse exerted on the tent peg by the hammer, stating the units of your answer. (3)

The ground exerts a constant vertical resistive force of magnitude R newtons, bringing the hammer and tent peg to rest after they travel a distance of 12 cm .

- (c) Find the value of R . (5)

October 2023 Q3 — detailed solutionScalar $F = ma$ **ALL**
10 marks**Use conservation of momentum**

Take downward as positive during the impact. Before impact, the hammer has momentum $1.8(10)$ and the peg has momentum 0. After impact, the combined mass is $1.8 + 0.2 = 2.0$ kg.

Find the common speed

$1.8(10) = 2.0v$, so $18 = 2v$ and $v = 9 \text{ m s}^{-1}$.

Use impulse on the peg

The peg changes from rest to speed 9 m s^{-1} downward. The impulse magnitude is $0.2(9 - 0) = 1.8 \text{ N s}$.

Find the deceleration in the ground

The hammer and peg move together from speed 9 to rest over 0.12 m. With downward positive, $0 = 9^2 + 2a(0.12)$, so $a = -337.5 \text{ m s}^{-2}$.

Apply Newtons second law

For the combined mass 2.0 kg, forces are weight $2g$ downward and resistance R upward. With downward positive, $2g - R = 2a$.

Find R

$19.6 - R = 2(-337.5) = -675$, so $R = 694.6 \text{ N}$, which is about 695 N.

Final answer (a) $v = 9 \text{ m s}^{-1}$. (b) $I = 1.8 \text{ N s}$. (c) $R \approx 695 \text{ N}$.

Summer 2025 Q4 — complete original question

4.

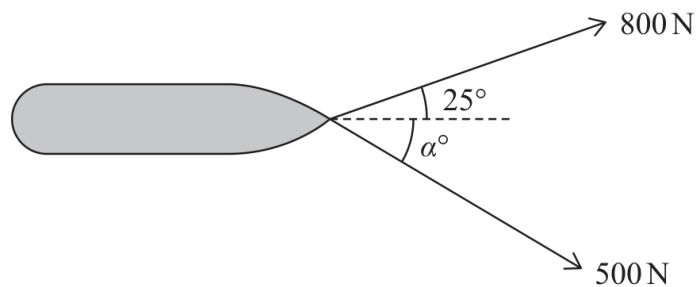


Figure 2

Two ropes are attached to a point on the front of a barge.
The barge is being pulled horizontally in a straight line along the centre of a long straight canal.

One rope makes an angle of 25° with the direction of motion of the barge and has a tension of 800 N.

The other rope makes an angle of α° with the direction of motion of the barge and has a tension of 500 N, as shown in Figure 2.

Both ropes are horizontal.

(a) Find the value of α

(3)

The mass of the barge is 15 tonnes and the resistance to the motion of the barge is a constant force of magnitude 750 N.

(b) Find the acceleration of the barge.

(4)

Summer 2025 Q4 — detailed solutionScalar $F = ma$ **ALL**
7 marks**Part (a): Find the value of α - Resolve perpendicular to direction of motion**

$$\sum F_{\perp} = 0 \text{ (no sideways motion)}$$

$$\text{Perpendicular component of 800 N rope: } 800 \sin 25^{\circ} = 800 \times 0.4226 = 338.1 \text{ N}$$

$$\text{Perpendicular component of 500 N rope: } 500 \sin \alpha$$

Part (a): Find the value of α - Set components equal

$$800 \sin 25^{\circ} = 500 \sin \alpha \quad 338.1 = 500 \sin \alpha \quad \sin \alpha = \frac{338.1}{500} \quad \sin \alpha = 0.6762 \quad \alpha = \sin^{-1}(0.6762) \quad \alpha = 42.5^{\circ}$$

Part (b): Find acceleration of barge - Resolve parallel to direction of motion

$$\sum F_{\parallel} = ma$$

$$\text{Parallel component of 800 N rope: } 800 \cos 25^{\circ} = 800 \times 0.9063 = 725.0 \text{ N}$$

$$\text{Parallel component of 500 N rope: } 500 \cos 42.5^{\circ} = 500 \times 0.7373 = 368.7 \text{ N}$$

$$\text{Resistance force: } 750 \text{ N (opposes motion)}$$

$$\text{Mass: } 15 \text{ tonnes} = 15000 \text{ kg}$$

Part (b): Find acceleration of barge - Apply Newton's Second Law

$$725.0 + 368.7 - 750 = 15000 \times a \quad 343.7 = 15000a \quad a = \frac{343.7}{15000} \quad a = 0.02291... \quad a = 0.023 \text{ m s}^{-2} \text{ (2 sf)}$$

Final answer (a) $\alpha = 43^{\circ}$ to the nearest degree. (b) $a = 0.023 \text{ m s}^{-2}$.

Summer 2021 Q4 — complete original question

4.

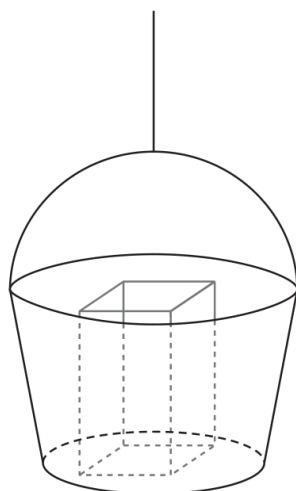


Figure 1

Figure 1 shows a large bucket used by a crane on a building site to move materials between the ground and the top of the building. The mass of the bucket is 15 kg.

The bucket is attached to a vertical cable with the bottom of the bucket horizontal. The cable is modelled as light and inextensible.

When the bucket is on the ground, a bag of cement of mass 25 kg is placed in the bucket.

The bucket with the bag of cement moves vertically upwards with constant acceleration 0.2 m s^{-2} . Air resistance is modelled as being negligible.

(a) Find the tension in the cable.

(3)

At the top of the building, the bag of cement is removed. A box of tools of mass 12 kg is now placed in the bucket.

Later on the bucket with the box of tools is moving vertically downwards with constant deceleration 0.1 m s^{-2} . Air resistance is again modelled as being negligible.

(b) Find the magnitude of the normal reaction between the bucket and the box of tools.

(3)

Summer 2021 Q4 — detailed solution

Lift or apparent reaction

ALL
6 marks**Find total mass going up**

The bucket and cement have total mass $15 + 25 = 40$ kg and acceleration 0.2 m s^{-2} upward.

Apply Newtons second law

Taking upward as positive: $T - 40g = 40(0.2)$.

Find the tension

$T - 392 = 8$, so $T = 400$ N.

Interpret the downward deceleration

The bucket and tools are moving downward but decelerating, so their acceleration is upward with magnitude 0.1 m s^{-2} .

Consider the box of tools

For the 12 kg box, upward force is the normal reaction R , and downward force is weight $12g$.

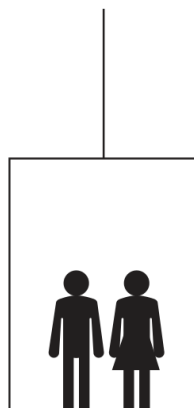
Find R

$R - 12g = 12(0.1)$. Thus $R = 117.6 + 1.2 = 118.8$ N, about 119 N.

Final answer (a) $T = 400$ N. (b) $R = 119$ N approximately.

October 2022 Q4 — complete original question

4.

**Figure 3**

Two children, Alan and Bhavana, are standing on the horizontal floor of a lift, as shown in Figure 3.

The lift has mass 250 kg. The lift is raised vertically upwards with constant acceleration by a vertical cable which is attached to the top of the lift. The cable is modelled as being light and inextensible. While the lift is accelerating upwards, the tension in the cable is 3616 N.

As the lift accelerates upwards, the floor of the lift exerts a force of magnitude 565 N on Alan and a force of magnitude 226 N on Bhavana.

Air resistance is modelled as being negligible and Alan and Bhavana are modelled as particles.

(a) By considering the forces acting on the lift only, find the acceleration of the lift. (3)

(b) Find the mass of Alan. (3)

October 2022 Q4 — detailed solution

Lift or apparent reaction

ALL
6 marks**Consider the lift only**

For the lift, the upward force is the cable tension 3616 N. Downward forces on the lift are its own weight $250g$ and the contact forces from Alan and Bhavana, 565 N and 226 N.

Apply Newtons second law

Taking upward as positive: $3616 - 250g - 565 - 226 = 250a$.

Find the acceleration

$3616 - 2450 - 565 - 226 = 250a$, so $375 = 250a$ and $a = 1.5 \text{ m s}^{-2}$.

Consider Alan

Let Alan's mass be m_A . The upward force on Alan is 565 N and his weight is m_Ag .

Find Alan's mass

$565 - m_Ag = m_A(1.5)$, so $565 = m_A(9.8 + 1.5) = 11.3m_A$. Hence $m_A = 50 \text{ kg}$.

Final answer (a) $a = 1.5 \text{ m s}^{-2}$. (b) mass of Alan = 50 kg.

January 2023 Q7 — complete original question

7.

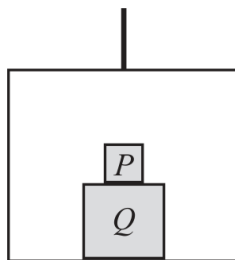


Figure 4

A simple lift operates by means of a vertical cable which is attached to the top of the lift.

The lift has mass m

A box Q is placed on the floor of the lift.

A box P is placed directly on top of box Q , as shown in Figure 4.

The cable is modelled as being light and inextensible and air resistance is modelled as being negligible.

The tension in the cable is $\frac{42mg}{5}$

The lift and its contents move vertically upwards with acceleration $\frac{2g}{5}$

Using the model,

- (a) find, in terms of m , the combined mass of boxes P and Q (4)

During the motion of the lift, the force exerted on box P by box Q is $\frac{14mg}{5}$

Using the model,

- (b) find, in terms of m , the mass of box P (3)

January 2023 Q7 — detailed solution

Lift or apparent reaction

ALL
7 marks**Define the box mass**

Let the combined mass of boxes P and Q be M . The lift itself has mass m , so the total moving mass is $m + M$.

Apply Newtons second law to the whole system

Taking upward as positive, $\frac{42}{5}mg - (m + M)g = (m + M)\frac{2g}{5}$.

Find M

Move the weight term to the right: $\frac{42}{5}mg = (m + M)\left(g + \frac{2g}{5}\right) = \frac{7g}{5}(m + M)$. Hence $42m = 7(m + M)$ and $M = 5m$.

Let the mass of P be p

For box P alone, the upward force from Q is $\frac{14}{5}mg$, while its weight is pg . The acceleration is still $\frac{2g}{5}$ upward.

Apply Newtons second law to P

$$\frac{14}{5}mg - pg = p\frac{2g}{5}.$$

Find p

Divide by g : $\frac{14}{5}m = p + \frac{2p}{5} = \frac{7p}{5}$. Therefore $p = 2m$.

Final answer (a) combined mass of P and Q = $5m$. (b) mass of P = $2m$.

TOPIC 07 | FAMILY 02

Connected-particle systems

WHAT THIS FAMILY ASKS

Model particles linked by light inextensible strings over smooth pulleys and solve coupled equations.

VARIANT ROUTE

V01 Horizontal-table particle connected to a hanging particle

V02 Incline connected to another particle

V03 Multiple strings or tension levels

V04 Two vertically hanging connected particles

V05 Limiting threshold followed by connected motion

13 classified questions

178 total marks

October 2022 Q7 — complete original question

7.

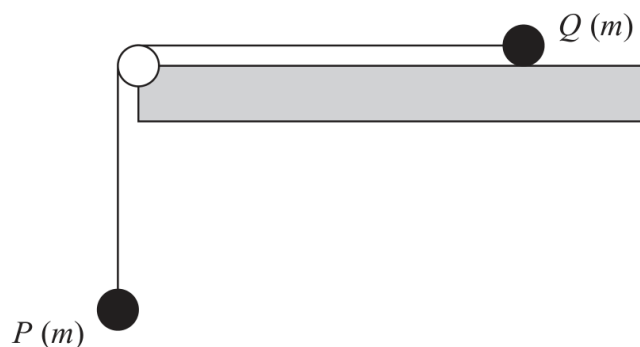


Figure 4

A particle P of mass m is attached to one end of a light inextensible string. Another particle Q , also of mass m , is attached to the other end of the string. The string passes over a small smooth pulley which is fixed at the edge of a rough horizontal table. Particle Q is held at rest on the table and particle P hangs vertically below the pulley with the string taut, as shown in Figure 4.

The pulley, P and Q all lie in the same vertical plane.

The coefficient of friction between Q and the table is μ , where $\mu < 1$

Particle Q is released from rest.

The tension in the string before Q hits the pulley is kmg , where k is a constant.

(a) Find k in terms of μ .

(7)

Given that Q is initially a distance d from the pulley,

(b) find, in terms of d , g and μ , the time taken by Q , after release, to reach the pulley.

(4)

(c) Describe what would happen if $\mu \geq 1$, giving a reason for your answer.

(2)

October 2022 Q7 — detailed solution

Horizontal table connected to a hanging particle

ALL
13 marks**Write the friction on Q**For Q on the horizontal table, $R = mg$, so the friction is $F = \mu mg$.**Apply Newtons second law to P**P moves downward and Q moves towards the pulley. For P, taking downward as positive: $mg - T = ma$.**Apply Newtons second law to Q**For Q, taking motion towards the pulley as positive: $T - \mu mg = ma$.**Use $T = kmg$** From P, $a = g(1 - k)$. From Q, $a = g(k - \mu)$.**Find k**Equate the two accelerations: $1 - k = k - \mu$. Hence $2k = 1 + \mu$ and $k = \frac{1 + \mu}{2}$.**Find the acceleration**

$$a = g(1 - k) = g\left(1 - \frac{1 + \mu}{2}\right) = \frac{g(1 - \mu)}{2}.$$

Use motion from restQ starts from rest and travels distance d , so $d = \frac{1}{2}at^2 = \frac{1}{2} \cdot \frac{g(1 - \mu)}{2}t^2$.**Find the time**

$$d = \frac{g(1 - \mu)t^2}{4}, \text{ so } t = \sqrt{\frac{4d}{g(1 - \mu)}}.$$

Consider mu at least 1If $\mu \geq 1$, the maximum available friction on Q is at least mg , so it can balance the pull caused by the hanging particle. The system would not start moving.

Final answer (a) $k = \frac{1 + \mu}{2}$. (b) $t = \sqrt{\frac{4d}{g(1 - \mu)}}$. (c) If $\mu \geq 1$, the system would not move because friction can balance the pull from P.

January 2021 Q8 — complete original question

8.

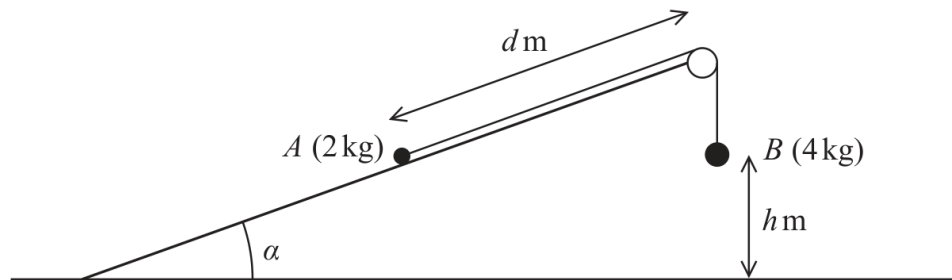


Figure 3

Two particles, A and B , have masses 2 kg and 4 kg respectively. The particles are connected by a light inextensible string. The string passes over a small smooth pulley which is fixed at the top of a rough plane. The plane is inclined to the horizontal ground at an angle α where $\tan \alpha = \frac{3}{4}$. The particle A is held at rest on the plane at a distance d metres from the pulley. The particle B hangs freely at rest, vertically below the pulley, at a distance h metres above the ground, as shown in Figure 3. The part of the string between A and the pulley is parallel to a line of greatest slope of the plane. The coefficient of friction between A and the plane is $\frac{1}{4}$.

The system is released from rest with the string taut and B descends.

- (a) Find the tension in the string as B descends.

(9)

On hitting the ground, B immediately comes to rest.

Given that A comes to rest before reaching the pulley,

- (b) find, in terms of h , the range of possible values of d .

(7)

- (c) State one physical factor, other than air resistance, that could be taken into account to make the model described above more realistic.

(1)

(Total 17 marks)

January 2021 Q8 — detailed solution

Incline connected to another particle

ALL
17 marks**Use the trig ratios**

Since $\tan \alpha = \frac{3}{4}$, $\sin \alpha = \frac{3}{5}$ and $\cos \alpha = \frac{4}{5}$.

Find friction on A

For A, $R = 2g \cos \alpha = \frac{8g}{5}$. With coefficient $\frac{1}{4}$, friction is $F = \frac{1}{4}R = \frac{2g}{5}$.

Equation for A

A moves up the plane. Taking up the plane as positive: $T - 2g \sin \alpha - F = 2a$.

Simplify A's equation

$T - 2g \cdot \frac{3}{5} - \frac{2g}{5} = 2a$, so $T - \frac{8g}{5} = 2a$.

Equation for B

B descends. Taking downward as positive: $4g - T = 4a$.

Solve for T

From A, $T = 2a + \frac{8g}{5}$. From B, $T = 4g - 4a$. Equating gives $6a = \frac{12g}{5}$, so $a = \frac{2g}{5}$. Then $T = 4g - 4\left(\frac{2g}{5}\right) = \frac{12g}{5}$ N.

Find speed when B reaches the ground

Before B hits the ground, both particles move distance h from rest with acceleration $\frac{2g}{5}$. Thus $v^2 = 2\left(\frac{2g}{5}\right)h = \frac{4gh}{5}$.

Find extra distance after B stops

After B hits the ground, A continues up the plane. The forces down the plane on A are $2g \sin \alpha + F = \frac{6g}{5} + \frac{2g}{5} = \frac{8g}{5}$, so its deceleration is $\frac{4g}{5}$.

Calculate extra distance

Let the extra distance be s . Then $0 = \frac{4gh}{5} - 2\left(\frac{4g}{5}\right)s$, so $s = \frac{h}{2}$.

Find the range for d

A travels total distance $h + \frac{h}{2} = \frac{3h}{2}$ before stopping. Since it comes to rest before reaching the pulley, $d > \frac{3h}{2}$.

Name a more realistic factor

A more realistic model could include friction at the pulley.

Final answer (a) $T = \frac{12g}{5} \text{ N} = 23.5 \text{ N}$. (b) $d > \frac{3h}{2}$. (c) A more realistic model could include friction at the pulley.

DR ESLAM AHMED

October 2021 Q7 — complete original question

7.

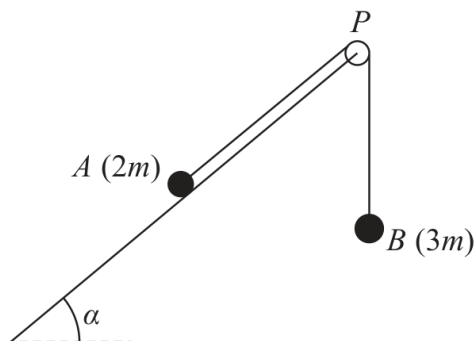


Figure 4

One end of a light inextensible string is attached to a particle A of mass $2m$. The other end of the string is attached to a particle B of mass $3m$. The string passes over a small, smooth, light pulley P which is fixed at the top of a rough inclined plane. The plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{3}{4}$

Particle A is held at rest on the plane with the string taut and B hanging freely below P , as shown in Figure 4. The section of the string AP is parallel to a line of greatest slope of the plane.

The coefficient of friction between A and the plane is $\frac{1}{2}$

Particle A is released and begins to move up the plane.

For the motion before A reaches the pulley,

(a) (i) write down an equation of motion for A ,

(ii) write down an equation of motion for B ,

(4)

(b) find, in terms of g , the acceleration of A ,

(5)

(c) find the magnitude of the force exerted on the pulley by the string.

(4)

(d) State how you have used the information that P is a smooth pulley.

(1)

October 2021 Q7 — detailed solution

Incline connected to another particle

ALL
14 marks**Equation for A**

A moves up the plane, so friction and the component of weight act down the plane. Taking up the plane as positive: $T - 2mg \sin \alpha - F = 2ma$.

Equation for B

B moves downward. Taking downward as positive: $3mg - T = 3ma$.

Find friction on A

Since $\tan \alpha = \frac{3}{4}$, $\sin \alpha = \frac{3}{5}$ and $\cos \alpha = \frac{4}{5}$. The normal reaction on A is $R = 2mg \cos \alpha = \frac{8mg}{5}$, so $F = \frac{1}{2}R = \frac{4mg}{5}$.

Simplify A's equation

$T - 2mg \cdot \frac{3}{5} - \frac{4mg}{5} = 2ma$, so $T - 2mg = 2ma$.

Solve with B's equation

From B, $T = 3mg - 3ma$. Substitute into $T - 2mg = 2ma$: $3mg - 3ma - 2mg = 2ma$. Thus $mg = 5ma$ and $a = \frac{g}{5}$.

Find the tension

Using $3mg - T = 3m\left(\frac{g}{5}\right)$ gives $T = 3mg - \frac{3mg}{5} = \frac{12mg}{5}$.

Find the pulley resultant

The two tensions are one vertical and one along the plane. Components of the resultant are $T \cos \alpha$ horizontally and $T + T \sin \alpha$ vertically.

Calculate the magnitude

Resultant $= T \sqrt{\cos^2 \alpha + (1 + \sin \alpha)^2} = T \sqrt{\left(\frac{4}{5}\right)^2 + \left(\frac{8}{5}\right)^2} = \frac{4\sqrt{5}}{5}T$. With $T = \frac{12mg}{5}$, this is $\frac{48\sqrt{5}}{25}mg$.

Use the smooth pulley information

The pulley is smooth, so the tension is the same on both sides of the pulley.

Final answer (a)(i) $T - 2mg \sin \alpha - F = 2ma$. (a)(ii) $3mg - T = 3ma$. (b) $a = \frac{g}{5}$. (c) resultant $= \frac{48\sqrt{5}}{25}mg$. (d) The tension is the same on both sides of the pulley.

January 2022 Q7 — complete original question

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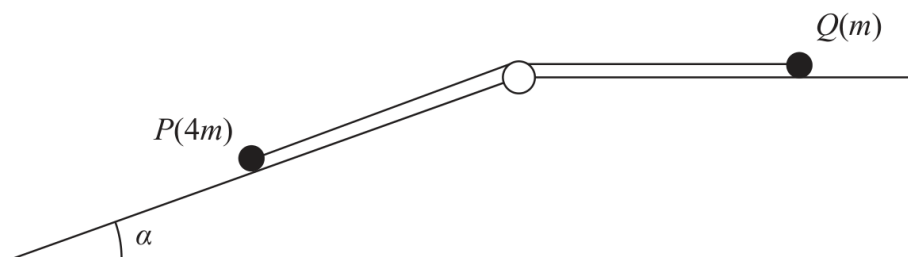


Figure 4

A particle P of mass $4m$ lies on the surface of a fixed rough inclined plane.

The plane is inclined to the horizontal at an angle α where $\tan \alpha = \frac{3}{4}$

The particle P is attached to one end of a light inextensible string.

The string passes over a small smooth pulley that is fixed at the top of the plane. The other end of the string is attached to a particle Q of mass m which lies on a smooth horizontal plane.

The string lies along the horizontal plane and in the vertical plane that contains the pulley and a line of greatest slope of the inclined plane.

The system is released from rest with the string taut, as shown in Figure 4, and P moves down the plane.

The coefficient of friction between P and the plane is $\frac{1}{4}$

For the motion before Q reaches the pulley

(a) write down an equation of motion for Q , (1)

(b) find, in terms of m and g , the tension in the string, (7)

(c) find the magnitude of the force exerted on the pulley by the string. (4)

(d) State where in your working you have used the information that the string is light. (1)

January 2022 Q7 — detailed solution

Incline connected to another particle

ALL
13 marks**Write the equation for Q**Q lies on a smooth horizontal plane. The only horizontal force on Q is the tension, so $T = ma$.**Use the trig ratios**Since $\tan \alpha = \frac{3}{4}$, $\sin \alpha = \frac{3}{5}$ and $\cos \alpha = \frac{4}{5}$.**Find friction on P**For P, $R = 4mg \cos \alpha = 4mg \cdot \frac{4}{5} = \frac{16mg}{5}$. With coefficient $\frac{1}{4}$, friction is $F = \frac{1}{4}R = \frac{4mg}{5}$.**Apply Newtons second law to P**P moves down the plane. Taking down the plane as positive: $4mg \sin \alpha - T - F = 4ma$.**Substitute values** $4mg \cdot \frac{3}{5} - T - \frac{4mg}{5} = 4ma$, so $\frac{8mg}{5} - T = 4ma$.**Use $T = ma$** From Q, $a = \frac{T}{m}$. Substitute into the equation for P: $\frac{8mg}{5} - T = 4T$.**Find T** $\frac{8mg}{5} = 5T$, so $T = \frac{8}{25}mg$.**Find the pulley resultant**The pulley is pulled by two tensions of magnitude T : one horizontal and one along the inclined string. Their resultant has components $T - T \cos \alpha$ horizontally and $T \sin \alpha$ vertically.**Calculate the resultant**Magnitude = $\sqrt{(T - T \cos \alpha)^2 + (T \sin \alpha)^2} = T \sqrt{(1 - \frac{4}{5})^2 + (\frac{3}{5})^2} = \frac{T}{5} \sqrt{10}$. With $T = \frac{8}{25}mg$, the resultant is $\frac{8\sqrt{10}}{125}mg$.**Use the light-string model**

The string is light, so the tension can be treated as the same on the P side and the Q side of the pulley.

Final answer (a) $T = ma$. (b) $T = \frac{8}{25}mg$. (c) resultant = $\frac{8\sqrt{10}}{125}mg$. (d) Same tension was used on both sides of the pulley.

October 2023 Q7 — complete original question

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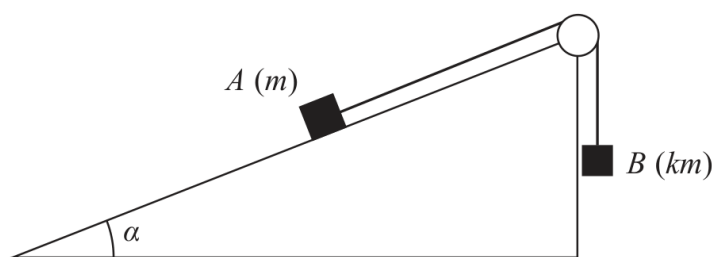


Figure 4

Figure 4 shows a block A of mass m held at rest on a rough plane. The plane is inclined at an angle α to the horizontal and the coefficient of friction between the block and the plane is μ .

One end of a light inextensible string is now attached to A . The string passes over a small smooth pulley which is fixed at the top of the plane.

The other end of the string is attached to a block B of mass km .

Block B hangs vertically below the pulley, with the string taut.

The string from A to the pulley lies along a line of greatest slope of the plane.

Both A and B are modelled as particles.

When the system is released from rest, A moves up the plane and the tension in the string is $\frac{4mg}{3}$

Given that $\mu = \frac{1}{3}$ and $\tan \alpha = \frac{7}{24}$

- (a) (i) find the magnitude of the acceleration of A , giving your answer in terms of g ,
 (ii) find the value of k .

(9)

- (b) Find the magnitude of the resultant force exerted on the pulley by the string, giving your answer in terms of m and g .

(4)

October 2023 Q7 — detailed solution

Incline connected to another particle

ALL
13 marks**Use the trig ratios**

Given $\tan \alpha = \frac{7}{24}$, use $\sin \alpha = \frac{7}{25}$ and $\cos \alpha = \frac{24}{25}$.

Find friction on A

For block A, $R = mg \cos \alpha = \frac{24mg}{25}$. Since $\mu = \frac{1}{3}$, friction is $F = \frac{1}{3}R = \frac{8mg}{25}$.

Apply Newtons second law to A

A moves up the plane, so friction and the component of weight act down the plane. Taking up the plane as positive: $T - mg \sin \alpha - F = ma$.

Find the acceleration

Substitute $T = \frac{4}{3}mg$, $mg \sin \alpha = \frac{7mg}{25}$ and $F = \frac{8mg}{25}$: $ma = \frac{4}{3}mg - \frac{15mg}{25} = \left(\frac{4}{3} - \frac{3}{5}\right)mg = \frac{11mg}{15}$.

Hence $a = \frac{11g}{15}$.

Apply Newtons second law to B

Block B moves downwards with the same acceleration. Taking downward as positive: $kmg - T = kma$.

Find k

Substitute $T = \frac{4}{3}mg$ and $a = \frac{11g}{15}$: $kmg - \frac{4}{3}mg = km \frac{11g}{15}$. Divide by mg to get $k - \frac{4}{3} = \frac{11k}{15}$, so $15k - 20 = 11k$ and $k = 5$.

Resolve the force on the pulley

The pulley is pulled by two tensions of magnitude T . One is vertical and the other is along the plane. The horizontal component of the resultant is $T \cos \alpha$ and the vertical component is $T + T \sin \alpha$.

Find the resultant

The magnitude is $T\sqrt{\cos^2 \alpha + (1 + \sin \alpha)^2}$. With $\sin \alpha = \frac{7}{25}$ and $\cos \alpha = \frac{24}{25}$, this is

$T\sqrt{\frac{576}{625} + \frac{1024}{625}} = \frac{8T}{5}$. Since $T = \frac{4}{3}mg$, the resultant is $\frac{32}{15}mg$.

Final answer (a)(i) $a = \frac{11g}{15}$. (a)(ii) $k = 5$. (b) resultant = $\frac{32}{15}mg$.

Summer 2024 Q8 — complete original question

8.

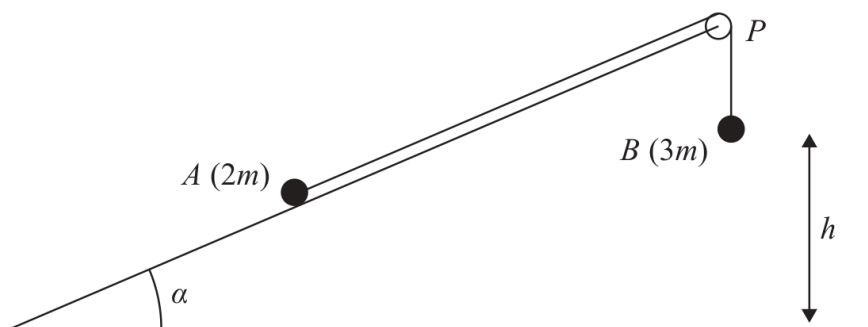


Figure 4

One end of a light inextensible string is attached to a particle A of mass $2m$. The other end of the string is attached to a particle B of mass $3m$. Particle A is held at rest on a rough plane which is inclined to horizontal ground at an angle α , where $\tan \alpha = \frac{5}{12}$.

The string passes over a small smooth pulley P which is fixed at the top of the plane. Particle B hangs vertically below P with the string taut, at a height h above the ground, as shown in Figure 4.

The part of the string between A and P lies along a line of greatest slope of the plane. The two particles, the string and the pulley all lie in the same vertical plane.

The coefficient of friction between A and the plane is $\frac{11}{36}$.

The particle A is released from rest and begins to move up the plane.

(a) Show that the frictional force acting on A as it moves up the plane is $\frac{22mg}{39}$. (3)

(b) Write down an equation of motion for B . (2)

(c) Show that the acceleration of A immediately after its release is $\frac{1}{3}g$. (4)

In the subsequent motion, A comes to rest before it reaches the pulley.

(d) Find, in terms of h , the total distance travelled by A from when it was released from rest to when it first comes to rest again. (6)

Summer 2024 Q8 — detailed solution

Incline connected to another particle

ALL
15 marks**Use the trig ratios**

Here $\tan \alpha = \frac{5}{12}$, so $\sin \alpha = \frac{5}{13}$ and $\cos \alpha = \frac{12}{13}$.

Show the frictional force on A

For A, $R = 2mg \cos \alpha = 2mg \cdot \frac{12}{13} = \frac{24mg}{13}$. Since $\mu = \frac{11}{36}$, $F = \mu R = \frac{11}{36} \cdot \frac{24mg}{13} = \frac{22mg}{39}$.

Write the equation for B

B moves down as A moves up. Taking downward as positive for B: $3mg - T = 3ma$.

Write the equation for A

Taking up the plane as positive for A: $T - 2mg \sin \alpha - F = 2ma$. Substituting $\sin \alpha = \frac{5}{13}$ and $F = \frac{22mg}{39}$ gives $T - \frac{10mg}{13} - \frac{22mg}{39} = 2ma$.

Show the acceleration

From B, $T = 3mg - 3ma$. Substitute into A equation: $3mg - 3ma - \frac{10mg}{13} - \frac{22mg}{39} = 2ma$. The constant part is $\frac{5mg}{3}$, so $\frac{5mg}{3} = 5ma$ and $a = \frac{g}{3}$.

Find the speed when B reaches the ground

While B descends distance h , A moves distance h up the plane from rest with acceleration $g/3$. Thus $v^2 = 2 \left(\frac{g}{3} \right) h = \frac{2gh}{3}$.

Find the later deceleration of A

After B reaches the ground, the string no longer pulls A. The forces down the plane on A are $2mg \sin \alpha + F = \frac{10mg}{13} + \frac{22mg}{39} = \frac{4mg}{3}$. Since A has mass $2m$, its deceleration is $\frac{2g}{3}$.

Find the extra distance and total distance

Let the extra distance be d . Using $0 = v^2 - 2 \left(\frac{2g}{3} \right) d$ with $v^2 = \frac{2gh}{3}$ gives $d = \frac{h}{2}$. Total distance travelled by A is $h + \frac{h}{2} = \frac{3h}{2}$.

Final answer (a) $F = \frac{22}{39}mg$. (b) $3mg - T = 3ma$. (c) $a = \frac{g}{3}$. (d) total distance = $\frac{3h}{2}$.

October 2025 Q7 — complete original question

7.

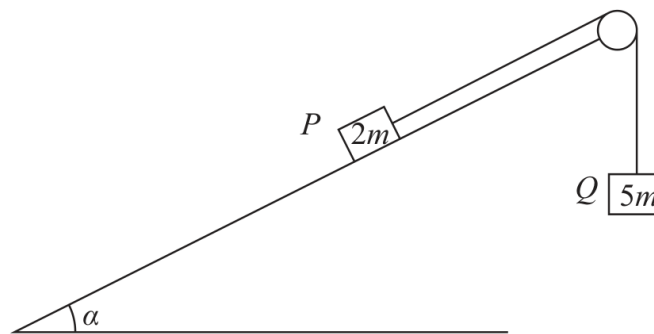


Figure 5

A block P of mass $2m$ is held at rest on a fixed rough plane.

The plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{5}{12}$

One end of a light inextensible string is attached to P .

The string is parallel to a line of greatest slope of the plane and passes over a smooth light pulley which is fixed at the top of the plane.

The other end of the string is attached to a block Q of mass $5m$.

Block Q hangs vertically below the pulley, as shown in Figure 5.

The system is released from rest with the string taut and block P moves up the plane.

Immediately after the system is released, the tension in the string is T and the acceleration of the blocks is a .

The blocks are modelled as particles and air resistance is ignored.

(a) Write down an equation of motion for Q .

(2)

The coefficient of friction between P and the plane is $\frac{1}{8}$

(b) Find T in terms of m and g .

(7)

(c) State how the solution to part (b) uses the fact that the string is inextensible.

(1)

The magnitude of the force exerted on the pulley by the string is kmg .

(d) Find the value of k to 3 significant figures.

(4)

October 2025 Q7 — detailed solution

Incline connected to another particle

ALL
14 marks**Part (a): Write equation of motion for Q**

Forces on Q (mass = 5m):

Taking downward as positive for Q:

For Q: $\sum F = ma$

$$5mg - T = 5ma$$

Part (b): Find T in terms of m and g - Resolve perpendicular to plane $\perp$ to plane: $\sum F = 0$

$$R = 2mg \cos \alpha \quad R = 2mg \times \frac{12}{13} \quad R = \frac{24mg}{13}$$

Part (b): Find T in terms of m and g - Find friction force

P is moving UP the plane, so friction acts DOWN the plane:

$$F = \mu R = \frac{1}{8} \times \frac{24mg}{13} = \frac{3mg}{13}$$

Part (b): Find T in terms of m and g - Resolve parallel to plane

Taking up the plane as positive:

 $\parallel$ to plane: $\sum F = ma$

$$T - 2mg \sin \alpha - F = 2ma \quad T - 2mg \times \frac{5}{13} - \frac{3mg}{13} = 2ma \quad T - \frac{10mg}{13} - \frac{3mg}{13} = 2ma \quad T - \frac{13mg}{13} = 2ma$$
$$T - mg = 2ma \quad \dots(1)$$

Part (b): Find T in terms of m and g - Combine with equation for QFrom part (a): $5mg - T = 5ma \dots(2)$

Add equations (1) and (2):

$$(T - mg) + (5mg - T) = 2ma + 5ma \quad 4mg = 7ma \quad a = \frac{4g}{7}$$

Part (b): Find T in terms of m and g - Substitute back to find T

From equation (1):

$$T = mg + 2ma = mg + 2m \times \frac{4g}{7} = mg + \frac{8mg}{7} = \frac{7mg + 8mg}{7} = \frac{15mg}{7}$$

Part (c): State how solution uses fact that string is inextensible

An inextensible string means the string cannot stretch.

This has the following consequence:

For an inextensible string over a pulley:

The magnitude of acceleration of both connected particles must be equal.

In this problem: $|a_P| = |a_Q| = a$

How this was used in part (b):

When we wrote the equations of motion:

; For P: $T - mg = 2ma$; For Q: $5mg - T = 5ma$ We used the same acceleration a for both particles because the string is inextensible.

If the string could stretch, the two particles could have different accelerations.

Part (d): Find value of k (force on pulley = kmg) - Identify the angle between tensionsThe string from Q is vertical (downward). The string from P is along the plane at angle α to horizontal.

Angle between the two strings = $90^\circ + \alpha$

Part (d): Find value of k (force on pulley = kmg) - Use vector components

Let's take horizontal (right) and vertical (up) as positive directions.

Tension from Q:

; Horizontal component: 0 ; Vertical component: $-T = -\frac{15mg}{7}$ (downward)

Tension from P along plane: The plane makes angle α with horizontal, so:

; Horizontal component: $-T \cos \alpha = -\frac{15mg}{7} \times \frac{12}{13} = -\frac{180mg}{91}$; Vertical component: $T \sin \alpha = \frac{15mg}{7} \times \frac{5}{13} = \frac{75mg}{91}$

Part (d): Find value of k (force on pulley = kmg) - Find resultant

$R_x = 0 - \frac{180mg}{91} = -\frac{180mg}{91}$ $R_y = -\frac{15mg}{7} + \frac{75mg}{91}$ Convert $\frac{15mg}{7}$ to denominator 91: $\frac{15mg}{7} = \frac{195mg}{91}$
 $R_y = -\frac{195mg}{91} + \frac{75mg}{91} = -\frac{120mg}{91}$

Part (d): Find value of k (force on pulley = kmg) - Find magnitude

$|R| = \sqrt{R_x^2 + R_y^2} = \sqrt{\left(\frac{180mg}{91}\right)^2 + \left(\frac{120mg}{91}\right)^2} = \frac{mg}{91} \sqrt{180^2 + 120^2} = \frac{mg}{91} \sqrt{32400 + 14400} = \frac{mg}{91} \sqrt{46800}$
 $= \frac{mg}{91} \times 216.333... = 2.377...mg = 2.38mg$ (3 s.f.) Therefore: $k = 2.38$

Final answer (a) $5mg - T = 5ma$. (b) $T = \frac{15mg}{7}$. (c) Both particles have the same acceleration magnitude. (d) $k = 2.38$.

Summer 2026 Q8 — complete original question

8.

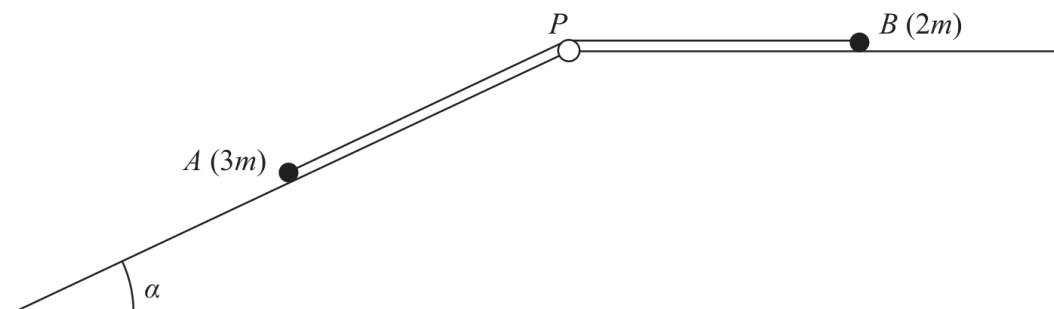


Figure 4

One end of a light inextensible string is attached to a particle A of mass $3m$. The other end of the string is attached to a particle B of mass $2m$. Particle A lies on a **rough** plane which is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{3}{4}$. Particle B lies on a **smooth** horizontal plane. The string passes over a small smooth pulley P which is fixed at the top of the inclined plane, as shown in Figure 4.

The two particles, the string and the pulley all lie in the same vertical plane. The portion of the string between A and P lies along a line of greatest slope of the inclined plane.

The coefficient of friction between A and the inclined plane is $\frac{1}{4}$.

The system is released from rest with the string taut and A begins to move down the inclined plane.

- (a) Find, in terms of m and g , the magnitude of the frictional force acting on A . (3)
- (b) Write down an equation of motion for B . (1)
- (c) Show that the tension in the string is $\frac{12mg}{25}$. (4)
- (d) State how you have used the fact that the string is inextensible. (1)
- The force exerted on the pulley by the string is S .
- (e) State clearly, giving your answer in terms of α , the direction of S . (1)
- (f) Find, in terms of m and g , the magnitude of S . (4)

Summer 2026 Q8 — detailed solution

Incline connected to another particle

ALL
14 marks**Part (a): direction and exact trigonometry**

From $\tan \alpha = 3/4$, use $\sin \alpha = 3/5$ and $\cos \alpha = 4/5$. Since A moves down the plane, friction acts up the plane.

Find the normal reaction and friction

Perpendicular to the plane, $R = 3mg \cos \alpha = 12mg/5$. Hence the friction magnitude is $\frac{1}{4}R = 3mg/5$.

Part (b): equation for B

Taking the direction toward the pulley as positive, the only horizontal force on B is the tension. Therefore $T = 2ma$.

Part (c): equation for A

Down the plane, $3mg \sin \alpha - T - 3mg/5 = 3ma$, so $6mg/5 - T = 3ma$.

Solve the simultaneous equations

Combine $6mg/5 - T = 3ma$ with $T = 2ma$. This gives $a = 6g/25$ and hence $T = 12mg/25$.

Part (d): use of inextensibility

The taut inextensible string makes A and B have the same acceleration magnitude, which is why the same a appears in both equations.

Part (e): direction of the pulley force

The pulley is pulled horizontally toward B and down the inclined branch toward A by equal tensions. Their resultant points downwards to the right at $90^\circ - \alpha/2$ below the horizontal.

Part (f): resultant magnitude

The angle between the tension forces is $180^\circ - \alpha$, so $S^2 = 2T^2(1 - \cos \alpha)$ and $S = 2T \sin(\alpha/2)$.

Substitute the tension

Using $\cos \alpha = 4/5$ gives $\sin(\alpha/2) = 1/\sqrt{10}$. With $T = 12mg/25$, $S = 12\sqrt{10}mg/125$.

Final answer (a) $\frac{3mg}{5}$. (b) $T = 2ma$. (c) $T = \frac{12mg}{25}$. (d) Same acceleration magnitude. (e) Downwards right at $90^\circ - \frac{\alpha}{2}$ below the horizontal. (f) $S = \frac{12\sqrt{10}mg}{125}$.

Summer 2024 Q3 — complete original question

3.

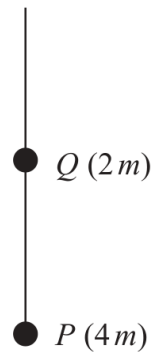


Figure 1

Two particles, P and Q , have masses $4m$ and $2m$ respectively. The particles are connected by a light inextensible string. A second light inextensible string has one end attached to Q . Both strings are taut and vertical, as shown in Figure 1.

The particles are **accelerating** vertically **downwards**.

Given that the tension in the string connecting the two particles is $3mg$, find, in terms of m and g , the tension in the upper string.

(6)

Summer 2024 Q3 — detailed solution

Multiple strings or tension levels

ALL
6 marks**Use the lower particle to find acceleration**

For P, take downward as positive. Its weight is $4mg$ and the tension in the string between P and Q is $3mg$ upward, so $4mg - 3mg = 4ma$.

Find the acceleration

The equation gives $mg = 4ma$, hence $a = \frac{g}{4}$ downward.

Write the equation for Q

For Q, downward forces are its weight $2mg$ and the lower string tension $3mg$. The upper string tension T acts upward. Taking downward as positive: $2mg + 3mg - T = 2ma$.

Substitute acceleration

$$5mg - T = 2m \left(\frac{g}{4} \right) = \frac{mg}{2}.$$

Find the upper tension

$$T = 5mg - \frac{mg}{2} = \frac{9mg}{2}.$$

Final answer $T = \frac{9mg}{2} = 4.5mg.$

October 2024 Q5 — complete original question

5.

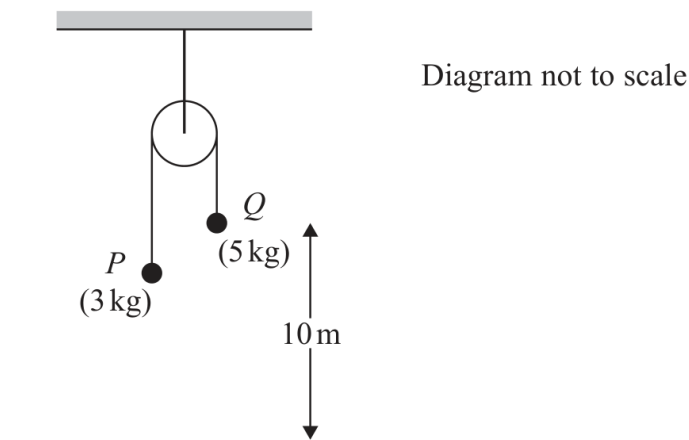


Figure 3

Two particles, P and Q , have masses 3 kg and 5 kg respectively. The particles are connected by a light inextensible string which passes over a small smooth fixed pulley.

The particles are released from rest with the string taut and the hanging parts of the string vertical, as shown in Figure 3.

Immediately after the particles are released from rest, P moves upwards with acceleration $a \text{ m s}^{-2}$ and the tension in the string is T newtons.

(a) Write down an equation of motion for P . (2)

(b) Find the value of T . (4)

The total force acting on the pulley due to the string has magnitude F newtons.

(c) Find the value of F . (2)

Initially, Q is 10 m above horizontal ground and P is more than 2 m below the pulley.

At the instant when Q has descended a distance of 2 m, the string breaks and Q falls to the ground.

(d) Find the speed of Q at the instant it hits the ground. (5)

October 2024 Q5 — detailed solution

Two vertically hanging connected particles

ALL
13 marks**Write the equation for P**P has mass 3 kg and moves upwards, so taking upwards as positive gives $T - 3g = 3a$.**Write the equation for Q**Q has mass 5 kg and moves downwards, so taking downwards as positive for Q gives $5g - T = 5a$.**Find acceleration and tension**Adding the two equations gives $2g = 8a$, so $a = \frac{g}{4}$. Substitute into $T - 3g = 3a$: $T = 3g + 3\left(\frac{g}{4}\right) = \frac{15g}{4} = 36.8 \text{ N}$.**Find the force on the pulley**The pulley is pulled by two vertical tensions, each of magnitude T . Hence $F = 2T = \frac{15g}{2} = 73.5 \text{ N}$.**Find Q speed when the string breaks**Before the string breaks, Q descends 2 m from rest with acceleration $g/4$. Thus $v^2 = 2\left(\frac{g}{4}\right)(2) = g$, so $v = \sqrt{g}$.**Find the remaining fall**After the string breaks, Q is still 8 m above the ground and accelerates freely under gravity. Let the impact speed be w . Then $w^2 = v^2 + 2g(8) = g + 16g = 17g$.**Calculate the impact speed** $w = \sqrt{17g} = \sqrt{166.6} = 12.9 \text{ m s}^{-1}$ to 3 significant figures.**Final answer** (a) $T - 3g = 3a$. (b) $T = 36.8 \text{ N}$. (c) $F = 73.5 \text{ N}$. (d) speed = 12.9 m s^{-1} .

January 2025 Q7 — complete original question

7.

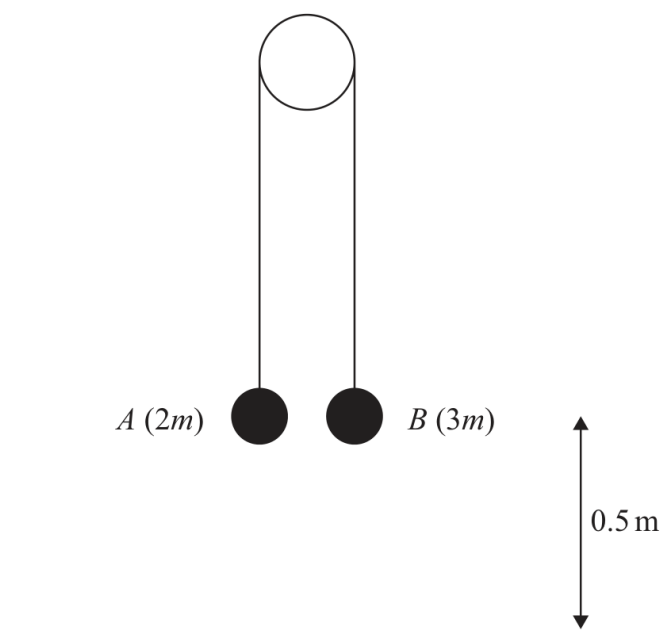


Figure 4

One end of a light inextensible string is attached to a particle A of mass $2m$.
 The other end is attached to a particle B of mass $3m$.
 The string passes over a small smooth fixed pulley.
 The string is taut and both straight parts of the string are vertical.
 Both particles are held at a distance 0.5 m above a horizontal surface, as shown in Figure 4.

The system is released from rest and B moves downwards.

(a) Find the tension in the string in terms of m and g . (5)

(b) Find the speed of B at the instant it strikes the surface. (4)

In the subsequent motion, A does not reach the pulley and B does not rebound after it strikes the surface.

(c) Find the time from the instant when the system is released from rest to the instant when A first reaches a height of 1.06 m above the surface. (6)

January 2025 Q7 — detailed solution

Two vertically hanging connected particles

ALL
15 marks**Write the equations of motion**

B moves down and A moves up with the same acceleration a . For A: $T - 2mg = 2ma$. For B: $3mg - T = 3ma$.

Find the acceleration

Adding the two equations gives $mg = 5ma$, so $a = \frac{g}{5}$.

Find the tension

Use $T - 2mg = 2ma$: $T = 2mg + 2m\left(\frac{g}{5}\right) = \frac{12mg}{5}$.

Find the speed when B hits the surface

B starts from rest and moves 0.5 m with acceleration $g/5$. Using $v^2 = u^2 + 2as$, $v^2 = 2\left(\frac{g}{5}\right)(0.5) = \frac{g}{5} = 1.96$.

Calculate the impact speed

$v = \sqrt{1.96} = 1.4 \text{ m s}^{-1}$.

Find the first time interval

The time until B hits the surface is found from $0.5 = \frac{1}{2}\left(\frac{g}{5}\right)t_1^2$. With $g = 9.8$, $t_1 = \frac{5}{7} = 0.714285 \dots$ s.

Find the extra time after B stops

After B hits the surface, A continues upwards freely under gravity. It is then at height 1.00 m and must rise another 0.06 m with initial speed 1.4 m s^{-1} . So $0.06 = 1.4t_2 - 4.9t_2^2$.

Find the total time

Solving $0.06 = 1.4t_2 - 4.9t_2^2$ gives the first positive time $t_2 = 0.0525 \dots$ s. Therefore the total time is $0.714285 \dots + 0.0525 \dots = 0.767$ s.

Final answer (a) $T = \frac{12mg}{5}$. (b) $v = 1.4 \text{ m s}^{-1}$. (c) $t = 0.767$ s approximately.

Summer 2025 Q8 — complete original question

8.

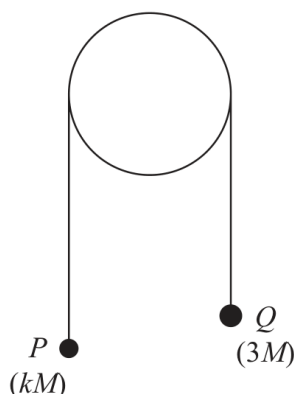


Figure 5

Two small balls, P and Q , have masses kM and $3M$ respectively, where $k < 3$

The balls are attached to the ends of a light inextensible string that passes over a fixed light smooth pulley.

The system is held at rest with the string taut and the hanging parts of the string vertical, as shown in Figure 5.

The system is released from rest and, in the subsequent motion, P moves with an acceleration of magnitude $\frac{1}{5}g$

The balls are modelled as particles.

(a) Write down an equation of motion for P .

(2)

(b) Find the value of k .

(3)

Given that $M = 0.5 \text{ kg}$,

(c) find the magnitude of the force exerted on the pulley by the string while Q is moving downwards.

(3)

At the instant when the system is released, P is more than 2.5 m from the pulley and Q is 2.5 m above horizontal ground.

After hitting the ground, Q rebounds with a speed of 0.4 m s^{-1}

(d) Find the magnitude of the impulse received by Q when it hits the ground.

(5)

In the subsequent motion, P does not hit the pulley.

(e) Find the total time from when the balls are released until P first comes to rest.

(4)

Summer 2025 Q8 — detailed solution

Two vertically hanging connected particles

ALL
17 marks**Part (a): Write equation of motion for P - Identify forces on P**

Particle P (mass kM) moves upward with acceleration $\frac{g}{5}$:
 ; Tension T (upward) ; Weight kMg (downward)

Part (a): Write equation of motion for P - Apply Newton's Second Law

Taking upward as positive:

$$\sum F = ma$$

$$T - kMg = kM \times \frac{g}{5}$$

Part (b): Find value of k - Write equation of motion for QParticle Q (mass $3M$) moves downward with acceleration $\frac{g}{5}$:

Taking downward as positive for Q:

$$3Mg - T = 3M \times \frac{g}{5} \quad 3Mg - T = \frac{3Mg}{5} \quad T = 3Mg - \frac{3Mg}{5} \quad T = \frac{12Mg}{5}$$

Part (b): Find value of k - Substitute into equation for PFrom part (a): $T - kMg = \frac{kMg}{5}$

$$\frac{12Mg}{5} - kMg = \frac{kMg}{5} \quad \frac{12Mg}{5} = kMg + \frac{kMg}{5} = \frac{5kMg + kMg}{5} = \frac{6kMg}{5} \quad 12 = 6k \quad k = 2$$

Part (c): Find force on pulley while Q moving down - Identify tension in stringFrom part (b): $T = \frac{12Mg}{5}$ Given: $M = 0.5 \text{ kg}$

$$T = \frac{12 \times 0.5 \times g}{5} = \frac{6g}{5} = \frac{6 \times 9.8}{5} = 11.76 \text{ N}$$

Part (c): Find force on pulley while Q moving down - Find force on pulley

The pulley experiences two tension forces:

; Horizontal tension from P: $T = 11.76 \text{ N}$; Vertical tension from Q: $T = 11.76 \text{ N}$

These are perpendicular, so use Pythagoras:

$$F = \sqrt{T^2 + T^2} = \sqrt{2T^2} = T\sqrt{2} = 11.76\sqrt{2} = 16.63... = 16.6 \text{ N (3 sf)}$$

$$\text{Or: } F = \frac{6g\sqrt{2}}{5} = \frac{12g}{5} = 2.4g = 23.5 \text{ N}$$

Wait, let me recalculate...

Actually: $F = 2T$ (both tensions pull on pulley)

$$F = 2 \times \frac{12Mg}{5} = \frac{24Mg}{5} = \frac{24 \times 0.5 \times 9.8}{5} = \frac{117.6}{5} = 23.5 \text{ N (3 sf)}$$

$$\text{Or: } \frac{12g}{5} = 23.5 \text{ N}$$

Part (d): Find impulse when Q hits ground - Find velocity of Q just before hitting ground

Using SUVAT with:

$$; u = 0 \text{ (starts from rest)} ; a = \frac{g}{5} = \frac{9.8}{5} = 1.96 \text{ m s}^{-2} ; s = 2.5 \text{ m}$$

$$v^2 = u^2 + 2as \quad v^2 = 0 + 2 \times \frac{g}{5} \times 2.5 \quad v^2 = \frac{5g}{5} \quad v^2 = g \quad v = \sqrt{g} = \sqrt{9.8} = 3.130... \text{ m s}^{-1}$$

$$\text{Or exact: } v = \sqrt{g} = \sqrt{\frac{7g}{5}}$$

$$\text{Actually: } v^2 = 2 \times 1.96 \times 2.5 = 9.8, \text{ so } v = \sqrt{9.8} = 3.13 \text{ m s}^{-1}$$

Part (d): Find impulse when Q hits ground - Calculate impulse

$$\text{Impulse} = m(v_{\text{after}} - v_{\text{before}})$$

Taking downward as positive before impact:

; Velocity before: $v_{\text{before}} = +3.13 \text{ m s}^{-1}$ (downward) ; Velocity after: $v_{\text{after}} = -0.4 \text{ m s}^{-1}$ (upward, rebounds) ; Mass of Q: $3M = 3 \times 0.5 = 1.5 \text{ kg}$

$$I = 1.5 \times (-0.4 - 3.13) = 1.5 \times (-3.53) = -5.295 \dots |I| = 5.3 \text{ N s (2 sf)}$$

$$\text{Or using exact values: } |I| = 1.5(\sqrt{g} + 0.4) = 1.5(3.13 + 0.4) = 5.3 \text{ N s}$$

Part (e): Find total time until P first comes to rest - Time for Q to fall (t_1)

Using $s = ut + \frac{1}{2}at^2$ with $u = 0$, $s = 2.5 \text{ m}$, $a = \frac{g}{5}$:

$$2.5 = 0 + \frac{1}{2} \times \frac{g}{5} \times t_1^2 \quad 2.5 = \frac{gt_1^2}{10} \quad 25 = gt_1^2 \quad t_1^2 = \frac{25}{9.8} = 2.551 \dots \quad t_1 = \sqrt{\frac{25}{9.8}} = 1.597 \dots \text{ s}$$

$$\text{Or exact: } t_1 = \sqrt{\frac{5 \times 5}{g}} = \frac{5}{\sqrt{g}}$$

$$\text{Better: } t_1 = \sqrt{\frac{5}{g/5}} = \sqrt{\frac{25}{g}}$$

$$\text{Actually: } 2.5 = \frac{1}{2} \times 1.96 \times t_1^2$$

$$t_1^2 = \frac{5}{1.96} = 2.551, \text{ so } t_1 = 1.597 \text{ s}$$

Part (e): Find total time until P first comes to rest - Velocity of P when Q hits ground

At time t_1 , P has velocity: $v_P = 0 + \frac{g}{5} \times t_1 = \frac{9.8}{5} \times 1.597 = 1.96 \times 1.597 = 3.13 \text{ m s}^{-1}$

Or: $v_P = \sqrt{g}$ (same as Q)

Part (e): Find total time until P first comes to rest - Time for P to come to rest (t_2)

After Q hits ground, string goes slack. P continues upward under gravity alone.

Using $v = u + at$ with $v = 0$, $u = 3.13 \text{ m s}^{-1}$, $a = -g$:

$$0 = 3.13 - 9.8t_2 \quad t_2 = \frac{3.13}{9.8} = 0.3194 \dots \text{ s}$$

$$\text{Or: } t_2 = \frac{\sqrt{g}}{g} = \frac{1}{\sqrt{g}}$$

Part (e): Find total time until P first comes to rest - Total time

$$T = t_1 + t_2 = 1.597 + 0.319 = 1.916 \dots = 1.9 \text{ s (2 sf)} \text{ or } 1.92 \text{ s (3 sf)}$$

Final answer (a) $T - kMg = \frac{kMg}{5}$. (b) $k = 2$. (c) $F = \frac{12g}{5} \text{ N} \approx 23.5 \text{ N}$. (d) $|I| = 5.30 \text{ N s}$. (e) $T = 1.92 \text{ s}$.

January 2026 Q7 — complete original question

7.

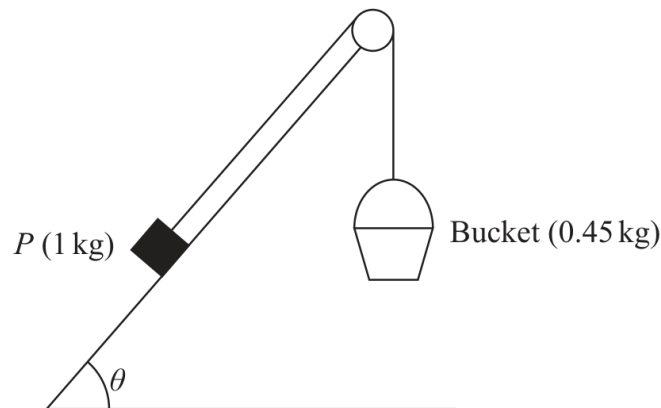


Figure 3

A rough plane is inclined at an angle θ to the horizontal, where $\tan \theta = \frac{4}{3}$

One end of a light inextensible string is attached to a package P .

The string passes over a smooth pulley that is fixed at the top of the plane.

The string from P to the pulley lies along a line of greatest slope of the plane.

The other end of the string is attached to a bucket.

The package P has mass 1 kg and is **held** at rest on the plane.

The bucket has mass 0.45 kg and hangs vertically below the pulley with the string taut, as shown in Figure 3.

The coefficient of friction between P and the plane is μ

When a small block of mass 0.2 kg is placed in the bucket, the system is released and remains at rest in **limiting equilibrium**, with P on the point of slipping down the plane.

- (a) Show that $\mu = \frac{1}{4}$ (7)

Additional identical blocks of mass 0.2 kg are added one at a time into the bucket until P starts to move **up** the plane.

At the instant when P starts to move up the plane, there are a **total** of n blocks in the bucket.

- (b) Find
- (i) the value of n
 - (ii) the magnitude of the initial acceleration of P
- (7)

January 2026 Q7 — detailed solution

Limiting threshold followed by connected motion

ALL
14 marks**Find the trigonometric ratios**

Since $\tan \theta = \frac{4}{3}$, use a 3-4-5 triangle: $\sin \theta = \frac{4}{5}$ and $\cos \theta = \frac{3}{5}$.

Use limiting equilibrium in part (a)

With one 0.2 kg block in the bucket, the bucket side has mass $0.45 + 0.2 = 0.65$ kg. Since the package is on the point of slipping down the plane, friction on P acts up the plane.

Find the friction force

The tension equals the bucket weight, so $T = 0.65g$. Along the plane for P in equilibrium, $T + F = g \sin \theta$. Hence $0.65g + F = \frac{4}{5}g$, so $F = \frac{3g}{20}$.

Find the normal reaction and μ

Perpendicular to the plane, $R = g \cos \theta = \frac{3}{5}g$. Since $F = \mu R$, $\mu = \frac{F}{R} = \frac{3g/20}{3g/5} = \frac{1}{4}$.

Set the condition for upward motion

When P is about to move up the plane, friction acts down the plane. With n blocks in the bucket, the bucket-side mass is $0.45 + 0.2n$. The bucket weight must exceed the downslope resistance: $(0.45 + 0.2n)g > g \sin \theta + \mu R$.

Find n

Using $g \sin \theta = \frac{4}{5}g$ and $\mu R = \frac{1}{4} \cdot \frac{3}{5}g = \frac{3g}{20}$, the condition is $0.45 + 0.2n > \frac{4}{5} + \frac{3}{20} = 0.95$. Hence $0.2n > 0.50$, so $n > 2.5$ and the least integer is $n = 3$.

Apply Newtons second law after motion begins

For $n = 3$, the bucket-side mass is $0.45 + 0.6 = 1.05$ kg and the total moving mass is $1 + 1.05 = 2.05$ kg. The driving force is $1.05g - \frac{4}{5}g - \frac{3g}{20} = 0.10g$.

Find the acceleration

Using $F = ma$, $0.10g = 2.05a$, so $a = \frac{0.10g}{2.05} = \frac{2g}{41} = 0.478 \text{ m s}^{-2}$ to 3 significant figures.

Final answer (a) $\mu = \frac{1}{4}$. (b)(i) $n = 3$. (b)(ii) $a = \frac{2g}{41} = 0.478 \text{ m s}^{-2}$ approximately.

TOPIC 07 | FAMILY 03

Towing and coupled vehicles

WHAT THIS FAMILY ASKS

Use whole-system and subsystem equations for tractors, trailers, tow bars or cables.

VARIANT ROUTE

V01 Whole system then one component

V02 Known tow force used to recover system dynamics

3 classified questions

25 total marks

Summer 2023 Q7 — complete original question

7.

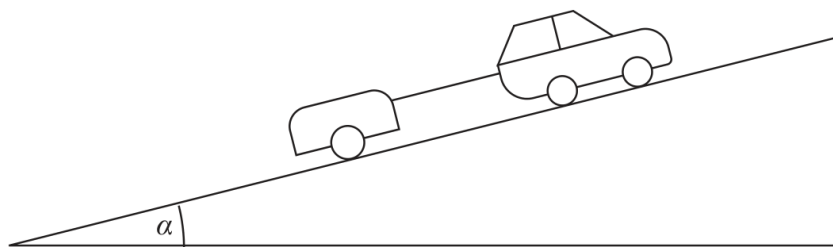


Figure 4

A car of mass 1200 kg is towing a trailer of mass 600 kg up a straight road, as shown in Figure 4.

The road is inclined at an angle α to the horizontal, where $\sin \alpha = \frac{1}{12}$

The driving force produced by the engine of the car is 3000 N.

The car moves with acceleration 0.75 m s^{-2}

The non-gravitational resistance to motion of

- the **car** is modelled as a constant force of magnitude $2R$ newtons
- the **trailer** is modelled as a constant force of magnitude R newtons

The car and the trailer are modelled as particles.

The tow bar between the car and trailer is modelled as a light rod that is parallel to the direction of motion.

Using the model,

(a) show that the value of R is 60

(4)

(b) find the tension in the tow bar.

(3)

When the car and trailer are moving at a speed of 12 m s^{-1} , the tow bar breaks.

Given that the non-gravitational resistance to motion of the trailer remains unchanged,

(c) use the model to find the further distance moved by the trailer before it first comes to rest.

(4)

Summer 2023 Q7 — detailed solution

Whole system then one component

ALL

11 marks

Use the whole system

For car plus trailer, the total mass is 1800 kg and the total resistance is $2R + R = 3R$. Taking up the slope as positive: $3000 - 1800g \sin \alpha - 3R = 1800(0.75)$.

Show R is 60

Since $\sin \alpha = \frac{1}{12}$, $1800g \sin \alpha = 1800 \cdot 9.8 \cdot \frac{1}{12} = 1470$. So $3000 - 1470 - 3R = 1350$, giving $180 = 3R$ and $R = 60$ N.

Apply Newtons second law to the trailer

For the trailer while it is being towed, $T - 600g \sin \alpha - R = 600(0.75)$.

Find the tow-bar tension

$T - 600 \cdot 9.8 \cdot \frac{1}{12} - 60 = 450$, so $T - 490 - 60 = 450$ and $T = 1000$ N.

Find the trailer acceleration after the break

After the tow bar breaks, the trailer is still moving up the slope, but the component of weight and resistance act down the slope. Taking up the slope as positive: $-600g \sin \alpha - 60 = 600a$.

Simplify the acceleration

$-490 - 60 = 600a$, so $a = -\frac{550}{600} = -\frac{11}{12} \text{ m s}^{-2}$.

Find the stopping distance

With initial speed 12 m s^{-1} and final speed 0, use $v^2 = u^2 + 2as$: $0 = 12^2 + 2 \left(-\frac{11}{12}\right) d$. Hence $d = \frac{864}{11} = 78.5 \dots \text{ m}$.

Final answer (a) $R = 60$ N. (b) $T = 1000$ N. (c) $d = 78.5$ m approximately.

October 2019 Q3 — complete original question

3. A car of mass 800 kg is towing a trailer of mass 400 kg up a straight road using a towbar. The towbar is parallel to the road and parallel to the direction of motion of the car. The road is inclined to the horizontal at an angle α , where $\sin \alpha = \frac{1}{7}$. The engine of the car produces a constant driving force of magnitude D newtons. The resistance to the motion of the car from non-gravitational forces is modelled as a single force of magnitude 420 N. The resistance to the motion of the trailer from non-gravitational forces is modelled as a single force of magnitude 300 N. The car and trailer are modelled as particles and the towbar is modelled as a light rod.

Given that the tension in the towbar is 2060 N, find the value of D .

(7)

DR ESLAM AHMED

October 2019 Q3 — detailed solution

Known tow force recovers system dynamics

ALL
7 marks**Use the trailer to find acceleration**

For the trailer, take up the road as positive. The forces along the road are tension 2060 N up the road, resistance 300 N down the road, and weight component $400g \sin \alpha$ down the road.

Form the trailer equation

$2060 - 300 - 400g \sin \alpha = 400a$. Since $\sin \alpha = \frac{1}{7}$, this becomes $2060 - 300 - 400g \cdot \frac{1}{7} = 400a$.

Find the acceleration

Using $g = 9.8$, $400g \cdot \frac{1}{7} = 560$. Hence $1200 = 400a$ and $a = 3 \text{ m s}^{-2}$.

Apply Newton's second law to the car

For the car: $D - 420 - 800g \sin \alpha - 2060 = 800a$.

Find the driving force

$D - 420 - 800g \cdot \frac{1}{7} - 2060 = 800(3)$. Since $800g \cdot \frac{1}{7} = 1120$, $D = 420 + 1120 + 2060 + 2400 = 6000 \text{ N}$.

Final answer $D = 6000 \text{ N}$.

Summer 2022 Q3 — complete original question

3. A tractor of mass 6 tonnes is dragging a large block of mass 2 tonnes along rough horizontal ground. The cable connecting the tractor to the block is horizontal and parallel to the direction of motion.

The cable is modelled as being light and inextensible.

The driving force of the tractor is 7400 N and the resistance to the motion of the tractor is 200 N. The resistance to the motion of the block is R newtons, where R is a constant.

Given that the tension in the cable is 6000 N and the tractor is accelerating,

- (a) find the value of R .

(6)

- (b) State how you have used the fact that the cable is modelled as being inextensible.

(1)

DR ESLAM AHMED

Summer 2022 Q3 — detailed solution

Known tow force recovers system dynamics

ALL
7 marks**Apply Newtons second law to the tractor**

The tractor has mass 6000 kg. Its forward driving force is 7400 N, and the backward forces are 200 N resistance and 6000 N tension.

Find the acceleration

$7400 - 200 - 6000 = 6000a$, so $1200 = 6000a$ and $a = 0.2 \text{ m s}^{-2}$.

Apply Newtons second law to the block

The block has mass 2000 kg. The forward force is the tension 6000 N and the resistance is R N.

Find R

$6000 - R = 2000(0.2) = 400$. Therefore $R = 5600$ N.

Use the inextensible cable model

The cable is inextensible, so the tractor and block move together with the same acceleration.

Final answer (a) $R = 5600$ N. (b) The tractor and block have the same acceleration.

TOPIC 07 | FAMILY 04

Rough horizontal dynamics

WHAT THIS FAMILY ASKS

Use reaction, friction and Newton's second law for motion on a rough horizontal surface, often after kinematics determines acceleration.

VARIANT ROUTE

V01 Infer horizontal friction from motion data

3 classified questions

40 total marks

January 2020 Q7 — complete original question

7.

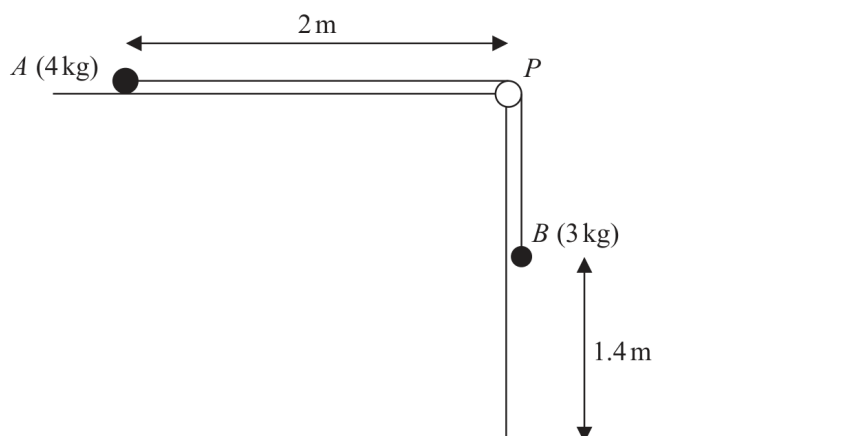


Figure 3

A particle A of mass 4 kg is held at rest on a rough horizontal table. Particle A is attached to one end of a string that passes over a pulley P . The pulley is fixed at the edge of the table. The other end of the string is attached to a particle B , of mass 3 kg , which hangs freely below P .

The part of the string from A to P is perpendicular to the edge of the table and A , P and B all lie in the same vertical plane.

The string is modelled as being light and inextensible and the pulley is modelled as being small, smooth and light.

The system is released from rest with the string taut. At the instant of release, A is 2 m from the edge of the table and B is 1.4 m above a horizontal floor, as shown in Figure 3.

After descending with constant acceleration for 2 seconds , B hits the floor and does not rebound.

(a) Show that the acceleration of A before B hits the floor is 0.7 m s^{-2} (2)

(b) State which of the modelling assumptions you have used in order to answer part (a). (1)

(c) Find the magnitude of the resultant force exerted on the pulley by the string. (4)

The coefficient of friction between A and the table is μ .

(d) Find the value of μ . (6)

(e) Determine, by calculation, whether or not A reaches the pulley. (5)

(Total 18 marks)

January 2020 Q7 — detailed solution

Infer horizontal friction from motion data

ALL
18 marks**Use B's descent**

Before B hits the floor, it descends 1.4 m from rest in 2 s. Use $s = ut + \frac{1}{2}at^2$: $1.4 = \frac{1}{2}a(2)^2$. Hence $a = 0.7 \text{ m s}^{-2}$.

Name the modelling assumption

The string is inextensible, so A and B have the same magnitude of acceleration while the string is taut.

Find the tension

For B, taking downward as positive: $3g - T = 3(0.7)$. Thus $T = 3g - 2.1 = 27.3 \text{ N}$.

Find the resultant force on the pulley

The pulley is pulled by two perpendicular tensions of magnitude T . Therefore the resultant is $\sqrt{T^2 + T^2} = T\sqrt{2} = 27.3\sqrt{2} = 38.6 \text{ N}$ approximately.

Find the friction on A

For A before B hits the floor, $T - F = 4(0.7)$. Hence $F = 27.3 - 2.8 = 24.5 \text{ N}$.

Find mu

On the horizontal table, $R = 4g = 39.2 \text{ N}$ and $F = \mu R$. So $\mu = \frac{24.5}{39.2} = 0.625$.

Find A's speed when B hits the floor

At this instant, $v = at = 0.7(2) = 1.4 \text{ m s}^{-1}$. A has also moved 1.4 m towards the pulley.

Find the extra stopping distance

After B hits the floor, A is slowed only by friction. Its deceleration is $\mu g = 0.625(9.8) = 6.125 \text{ m s}^{-2}$. Use $0 = (1.4)^2 - 2(6.125)s$, giving $s = 0.16 \text{ m}$.

Compare with the distance to the pulley

A's total distance is $1.4 + 0.16 = 1.56 \text{ m}$, which is less than 2 m. Therefore A does not reach the pulley.

Final answer (a) $a = 0.7 \text{ m s}^{-2}$. (b) Inextensibility of the string. (c) 38.6 N. (d) $\mu = 0.625$. (e) A does not reach the pulley.

October 2021 Q2 — complete original question

2. A particle P of mass $2m$ is moving on a rough horizontal plane when it collides directly with a particle Q of mass $4m$ which is at rest on the plane. The speed of P immediately before the collision is $3u$. The speed of Q immediately after the collision is $2u$.

(a) Find, in terms of u , the speed of P immediately after the collision.

(3)

(b) State clearly the direction of motion of P immediately after the collision.

(1)

Following the collision, Q comes to rest after travelling a distance $\frac{6u^2}{g}$ along the plane.

The coefficient of friction between Q and the plane is μ .

(c) Find the value of μ .

(6)

DR ESAM AHMED

October 2021 Q2 — detailed solution

Infer horizontal friction from motion data

ALL
10 marks**Use conservation of momentum**

Take P's original direction as positive. Before collision, P has velocity $3u$ and Q is at rest. After collision, Q has velocity $2u$. Let P's velocity be v .

Find P velocity

$2m(3u) = 2mv + 4m(2u)$. Thus $6mu = 2mv + 8mu$, so $2mv = -2mu$ and $v = -u$.

State speed and direction

The speed of P is u . The negative sign shows that P moves in the opposite direction after the collision.

Find the deceleration of Q

After collision, Q starts with speed $2u$ and stops after distance $\frac{6u^2}{g}$. Use $v^2 = u^2 + 2as$: $0 = (2u)^2 + 2a\left(\frac{6u^2}{g}\right)$.

Simplify acceleration

$0 = 4u^2 + \frac{12au^2}{g}$, so $a = -\frac{g}{3}$. The deceleration magnitude is $\frac{g}{3}$.

Use friction on Q

For Q, $R = 4mg$, so friction is $\mu R = 4\mu mg$. This friction produces deceleration: $4\mu mg = 4m \cdot \frac{g}{3}$.

Find μ

Cancel $4mg$: $\mu = \frac{1}{3}$.

Final answer (a) speed of P = u . (b) P reverses direction. (c) $\mu = \frac{1}{3}$.

January 2025 Q3 — complete original question

3.

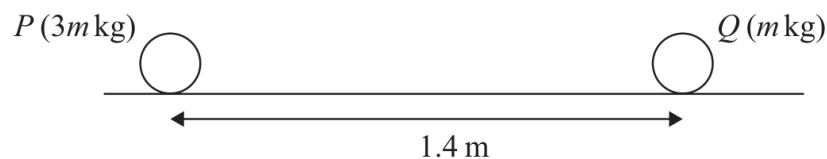


Figure 1

A particle P of mass $3m \text{ kg}$ and a particle Q of mass $m \text{ kg}$ are at rest on a rough horizontal surface. The distance between P and Q is 1.4 m , as shown in Figure 1.

An impulse of magnitude $\lambda \text{ N s}$ is now applied to P in the direction PQ . Immediately after the impulse is applied, the speed of P is 5 m s^{-1} .

(a) Find λ in terms of m .

(2)

Immediately before P collides with Q , the speed of P is 2.5 m s^{-1} . The coefficient of friction between P and the surface is μ .

(b) Find the value of μ .

(7)

Immediately after P collides with Q , the speed of Q is 2.1 m s^{-1} .

(c) Find the speed of P immediately after the collision.

(3)

January 2025 Q3 — detailed solution

Infer horizontal friction from motion data

ALL
12 marks**Use impulse to find lambda**

P starts from rest and immediately after the impulse has speed 5 m s^{-1} . Since the mass of P is $3m$,
 $\lambda = 3m(5 - 0) = 15m$.

Find the acceleration before collision

Between the impulse and the collision, P slows from 5 to 2.5 m s^{-1} over 1.4 m . Using $v^2 = u^2 + 2as$:
 $(2.5)^2 = 5^2 + 2a(1.4)$.

Calculate the acceleration

$$6.25 = 25 + 2.8a, \text{ so } a = \frac{-18.75}{2.8} = -6.696 \dots \text{ m s}^{-2}.$$

Connect friction to acceleration

On the horizontal surface, $R = 3mg$ and friction is $F = \mu R = 3\mu mg$. This friction is opposite the motion, so $-3\mu mg = 3ma$ and therefore $a = -\mu g$.

Find mu

$$\mu = \frac{-a}{g} = \frac{6.696 \dots}{9.8} = 0.683 \dots, \text{ so } \mu = 0.68 \text{ to 2 significant figures.}$$

Use momentum in the collision

Immediately before collision, P has speed 2.5 and Q is at rest. Immediately after, Q has speed 2.1 and P has speed v . Conservation of momentum gives $3m(2.5) + m(0) = 3mv + m(2.1)$.

Find the speed after collision

$$7.5 = 3v + 2.1, \text{ so } 3v = 5.4 \text{ and } v = 1.8 \text{ m s}^{-1}.$$

Final answer (a) $\lambda = 15m$. (b) $\mu = 0.68$ or 0.683 . (c) speed of P = 1.8 m s^{-1} .